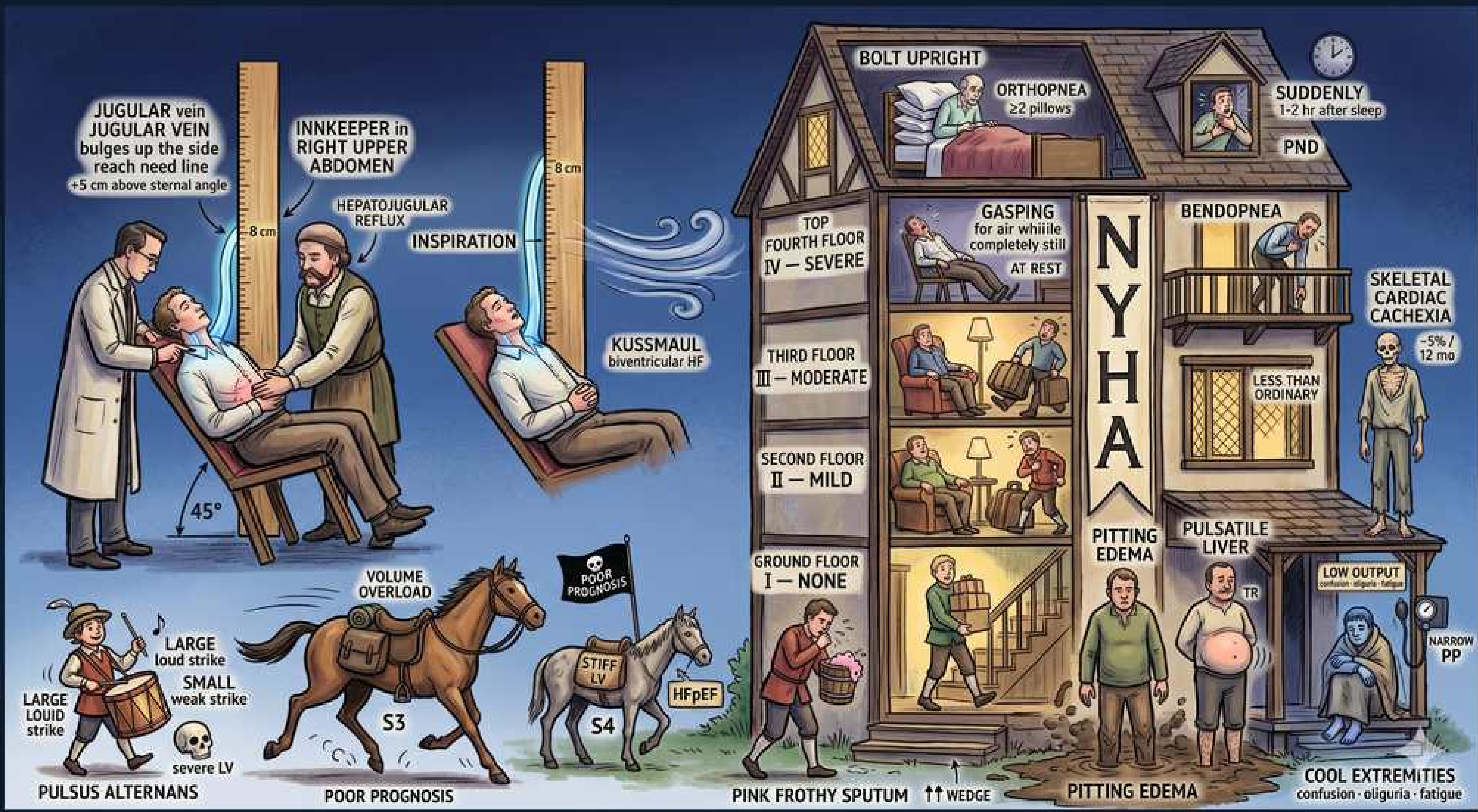
Heart Failure — SCENE 2 — HF EVALUATION & NYHA



1. JVP / elevated JVD

WHAT YOU SEE

Innkeeper pointing to bulging jugular vein on the neck

WHAT IT ENCODES

Elevated JVP reflects raised right atrial / central venous pressure. Measure as vertical height above the sternal angle: normal is ≤3 cm, so >3 cm above the angle (add 5 cm for distance to the RA = >8 cm H2O) is abnormal. The internal jugular is preferred (no valves, biphasic waveform that varies with respiration/position); examine at 45°. Elevated JVP is the single most useful bedside sign of volume overload and predicts HF hospitalization and death.

BOARD TRAP

A board stem will show a dyspneic patient and ask you to confirm a 'cardiac' cause vs. pulmonary disease — the wrong move is to call elevated JVD nonspecific and order CT chest first. Discriminator: elevated JVP plus an S3 has a high likelihood ratio for elevated filling pressures and HF; isolated COPD/pulmonary HTN can elevate JVP too, but the combination with orthopnea, displaced PMI, and S3 points to LV failure.

2. Hepatojugular reflux

WHAT YOU SEE

Hand pressing right upper abdomen, JVP rises on inspiration

WHAT IT ENCODES

Hepatojugular (abdominojugular) reflux: apply firm sustained pressure over the RUQ/mid-abdomen for 10–15 seconds; a sustained JVP rise of >3 cm (or >4 cm) that persists during compression is positive, indicating the right heart cannot accommodate the augmented venous return = volume overload / RV dysfunction / elevated PCWP. A normal heart shows only a transient rise. It correlates with PCWP >15 mmHg and supports a cardiac cause of dyspnea.

BOARD TRAP

Trap answer: 'a positive hepatojugular reflux diagnoses tricuspid regurgitation.' Wrong — TR classically produces a pulsatile liver and large CV waves, not specifically a positive HJR. A sustained positive HJR is a marker of high right-sided/biventricular filling pressures and is most useful for confirming a cardiac (vs pulmonary) cause of dyspnea, not for diagnosing a specific valve lesion.

3. Kussmaul sign

WHAT YOU SEE

Sign 'KUSSMAUL' with JVP rising on inspiration

WHAT IT ENCODES

Kussmaul sign = paradoxical RISE (or failure to fall) of the JVP during inspiration. Normally inspiration drops intrathoracic pressure and JVP falls; when the right heart cannot accept augmented venous return, JVP rises instead. Classic for constrictive pericarditis; also seen in restrictive cardiomyopathy, RV infarction, severe RV failure, and tricuspid stenosis.

BOARD TRAP

High-yield trap: Kussmaul sign is offered as the answer for cardiac tamponade. WRONG — tamponade gives pulsus paradoxus (≥10 mmHg inspiratory drop in systolic BP) and is typically Kussmaul-NEGATIVE. Kussmaul sign is the discriminator that points you toward constrictive pericarditis (or RV infarct) rather than tamponade.

4. S3 gallop = poor prognosis

WHAT YOU SEE

Galloping horse labeled S3 with a skull/poor-prognosis flag

WHAT IT ENCODES

S3 (ventricular gallop, 'Ken-tuc-ky') is a low-pitched early-diastolic sound from rapid passive ventricular filling into a volume-overloaded/dilated ventricle; best heard at the apex in left lateral decubitus with the bell. In an adult it strongly suggests elevated LV filling pressure and systolic HF (HFrEF); it is one of the highest specificity bedside signs and an independent predictor of poor prognosis, hospitalization, and death. (A physiologic S3 can be normal in children, athletes, and pregnancy.)

BOARD TRAP

Trap: the stem describes an S3 and asks you to label the patient as HFpEF/diastolic dysfunction. WRONG — S3 is the VOLUME/systolic sound (HFrEF), whereas S4 is the stiff-ventricle sound of HFpEF/diastolic dysfunction. Discriminator: S3 = rapid filling into a dilated volume-overloaded LV (poor prognosis marker); S4 = atrial kick into a non-compliant ventricle and is absent in atrial fibrillation.

5. S4 = HFpEF (stiff)

WHAT YOU SEE

Smaller horse labeled S4 next to 'STIFF / HFpEF' tag

WHAT IT ENCODES

S4 (atrial gallop, 'Ten-nes-see') is a late-diastolic sound produced when the atrium contracts ('atrial kick') against a stiff, non-compliant ventricle — the hallmark of diastolic dysfunction/HFpEF, LVH (HTN, aortic stenosis), and ischemia. Best heard at the apex with the bell. Because it requires organized atrial contraction, an S4 CANNOT exist in atrial fibrillation.

BOARD TRAP

Trap answer: an S4 is heard in a patient who is in atrial fibrillation. WRONG/impossible — S4 depends on the atrial kick, which is lost in AF. If a stem gives AF, eliminate any choice invoking S4. Conversely, an S4 in a hypertensive elderly woman with preserved EF supports HFpEF/diastolic dysfunction rather than systolic HF.

6. Pulsus alternans

WHAT YOU SEE

Drummer with alternating LARGE/SMALL beats, 'severe LV'

WHAT IT ENCODES

Pulsus alternans = regular rhythm with alternating strong and weak pulse amplitude (and alternating systolic BP) beat-to-beat, due to cyclic variation in stroke volume from severely depressed LV systolic function. Best detected by slowly deflating a BP cuff (Korotkoff sounds appear in alternate beats). It signals advanced/severe HFrEF and carries a poor prognosis.

BOARD TRAP

Trap: confusing pulsus alternans with pulsus paradoxus. Pulsus alternans is alternating amplitude at a REGULAR rhythm and indicates severe LV systolic dysfunction; pulsus paradoxus is an exaggerated (>10 mmHg) inspiratory FALL in systolic BP and points to tamponade, severe asthma/COPD, or constriction. Don't pick pulsus alternans for a tamponade or asthma stem.

7. NYHA I — no symptoms

WHAT YOU SEE

Ground floor, person resting comfortably, 'I — NONE'

WHAT IT ENCODES

NYHA Class I: no limitation of physical activity; ordinary activity causes no dyspnea, fatigue, or palpitations. NYHA is a FUNCTIONAL classification of current symptom burden and can move up or down with treatment, unlike ACC/AHA stages which only progress. A Stage C patient (structural disease + prior symptoms) who is now asymptomatic on GDMT can be NYHA I.

BOARD TRAP

Trap: equating NYHA I with ACC/AHA Stage A. WRONG — Stage A is at-risk with NO structural disease and NO symptoms (e.g., HTN/diabetes), whereas NYHA I requires established HF (structural disease, i.e., Stage C) that is currently asymptomatic. Discriminator: NYHA describes symptoms in patients who already have HF; ACC stages describe the disease continuum including pre-HF risk.

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8. NYHA II — mild

WHAT YOU SEE

2nd floor, 'II — MILD / slight limitation'

WHAT IT ENCODES

NYHA Class II: slight limitation of physical activity; comfortable at rest, but ORDINARY activity (e.g., climbing stairs, brisk walking) produces dyspnea, fatigue, or palpitations. NYHA class II–IV is a key gate for therapies: MRAs (spironolactone/eplerenone) are indicated in NYHA II–IV HFrEF, and ARNI/BB/SGLT2i benefit across symptomatic classes.

BOARD TRAP

Trap: a stem with a NYHA II HFrEF patient and the choice 'MRA not yet indicated, wait for class III–IV.' WRONG — MRA mortality benefit (EMPHASIS-HF) is established starting at NYHA II. Don't withhold an MRA in mildly symptomatic HFrEF as long as K+ <5.0 and eGFR >30.

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9. NYHA III — moderate

WHAT YOU SEE

3rd floor, 'III — MODERATE / less than ordinary'

WHAT IT ENCODES

NYHA Class III: marked limitation; comfortable at rest but LESS-than-ordinary activity (e.g., walking short distances, dressing) causes symptoms. NYHA III (often subdivided IIIa/IIIb) is the threshold for device therapy consideration: CRT for LVEF ≤35% with LBBB and QRS ≥150 ms on GDMT, and ICD for primary prevention in symptomatic HFrEF.

BOARD TRAP

Trap: offering CRT for any symptomatic HFrEF patient regardless of QRS. WRONG — CRT benefit is greatest with a wide QRS (≥150 ms) and LBBB morphology; a narrow QRS predicts no benefit (and possible harm, as in EchoCRT). Discriminator: it's the LBBB + wide QRS, not the NYHA class alone, that drives CRT selection.

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10. NYHA IV — severe (at rest)

WHAT YOU SEE

Top floor, person gasping for air, 'IV — SEVERE / at rest'

WHAT IT ENCODES

NYHA Class IV: symptoms at REST and inability to carry out any physical activity without discomfort. Advanced/refractory NYHA IV (Stage D) triggers evaluation for advanced therapies: inotropes (bridge/palliative), LVAD, transplant, or palliative/hospice care. ARNI initiation is cautioned in decompensated NYHA IV/hypotension.

BOARD TRAP

Trap: 'NYHA staging is irreversible/progressive like the ACC/AHA stages.' WRONG — NYHA class FLUCTUATES; a decompensated NYHA IV patient can improve to NYHA II after diuresis and GDMT. ONLY the ACC/AHA stages (A→B→C→D) are one-directional and never regress. This A-vs-B contrast (functional/reversible vs. structural/irreversible) is the classic exam discriminator.

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11. Orthopnea / PND

WHAT YOU SEE

Pillows 'ORTHOPNEA ≥2 pillows' and clock 'PND 1-2h after sleep'

WHAT IT ENCODES

Orthopnea = dyspnea on lying flat, relieved by sitting up (quantified by number of pillows, e.g., ≥2–3); due to redistribution of splanchnic/lower-extremity blood to the central circulation increasing pulmonary venous pressure. PND = paroxysmal nocturnal dyspnea, waking abruptly gasping 1–2 hours after falling asleep, relieved by standing/opening a window. Both are specific symptoms of elevated LV filling pressure/pulmonary congestion.

BOARD TRAP

Trap: attributing PND-like nocturnal awakening with cough/wheeze to asthma and ordering an inhaler. Discriminator: cardiac PND is relieved by sitting/standing within minutes and accompanies orthopnea, elevated JVP, and an S3; nocturnal asthma is relieved by bronchodilators and lacks congestion signs. 'Cardiac asthma' (wheeze from pulmonary edema) can mimic asthma — treat the failure, not just the bronchospasm.

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12. Pink frothy sputum / pulm edema

WHAT YOU SEE

Person coughing pink frothy sputum, '↑↑ wedge'

WHAT IT ENCODES

Pink frothy sputum + markedly elevated PCWP (>18, often >25 mmHg) = acute cardiogenic pulmonary edema from acute LV failure/flash edema. Exam shows bilateral crackles, S3, hypoxemia, and CXR with cephalization, Kerley B lines, perihilar 'bat-wing' edema, and effusions. Treat with sit upright + oxygen/NIV, IV loop diuretic, and IV nitrates/vasodilators if BP allows (afterload reduction).

BOARD TRAP

Trap: a hypertensive flash pulmonary edema stem where the chosen answer is large-volume IV fluids or beta-blocker uptitration. WRONG — acute decompensated/flash edema needs PRELOAD and AFTERLOAD reduction (nitrates, diuretic, NIV); do NOT initiate or uptitrate a beta-blocker during acute decompensation and do not fluid-load. Discriminator: pink frothy sputum + high wedge = congestion, not hypovolemia.

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13. Pitting edema / pulsatile liver

WHAT YOU SEE

Swollen legs 'PITTING EDEMA', 'PULSATILE LIVER'

WHAT IT ENCODES

Dependent pitting edema, ascites, and a tender/pulsatile hepatomegaly with elevated JVP indicate RIGHT-sided / systemic venous congestion (and functional tricuspid regurgitation when the liver is pulsatile). Chronic passive congestion causes 'cardiac cirrhosis'/congestive hepatopathy with elevated bilirubin and transaminases. The most common cause of right heart failure is LEFT heart failure.

BOARD TRAP

Trap: a patient with isolated lower-extremity pitting edema labeled as HF and started on a loop diuretic. WRONG without corroboration — bilateral leg edema is nonspecific (venous insufficiency, nephrotic syndrome, cirrhosis, CCB, hypoalbuminemia). Discriminator: HF edema is accompanied by ELEVATED JVP/positive HJR; if the JVP is normal, look for a non-cardiac cause before diuresing.

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14. Cool extremities / low output

WHAT YOU SEE

Bluish hands, 'COOL EXTREMITIES / confusion oliguria / LOW OUTPUT'

WHAT IT ENCODES

Cool, mottled extremities, narrow pulse pressure, oliguria, and confusion = the 'cold' low-output/hypoperfusion profile of advanced HF (Stevenson profiles: warm/cold × wet/dry; 'cold & wet' is the worst). Cold + congested signals consideration of inotropes (dobutamine/milrinone), and if hypotensive with end-organ hypoperfusion, cardiogenic shock — needs ICU, possibly mechanical support.

BOARD TRAP

Trap: in a 'cold & wet' low-output patient with marginal BP, the chosen answer is to aggressively diurese alone or to start/uptitrate a beta-blocker. WRONG — pure aggressive diuresis can drop output further and beta-blockers are contraindicated in acute low-output/cardiogenic shock; these patients often need inotropic support first. Discriminator: cool extremities + narrow pulse pressure + oliguria = LOW OUTPUT, manage perfusion before (or alongside) decongestion.
